

Hallucination or Adaptation?: A Context-Aware Evaluation Framework for LLM Confabulation

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Same LLM output deviates from established facts — *hallucination or adaptation?* → The AC framework lets context determine the answer.

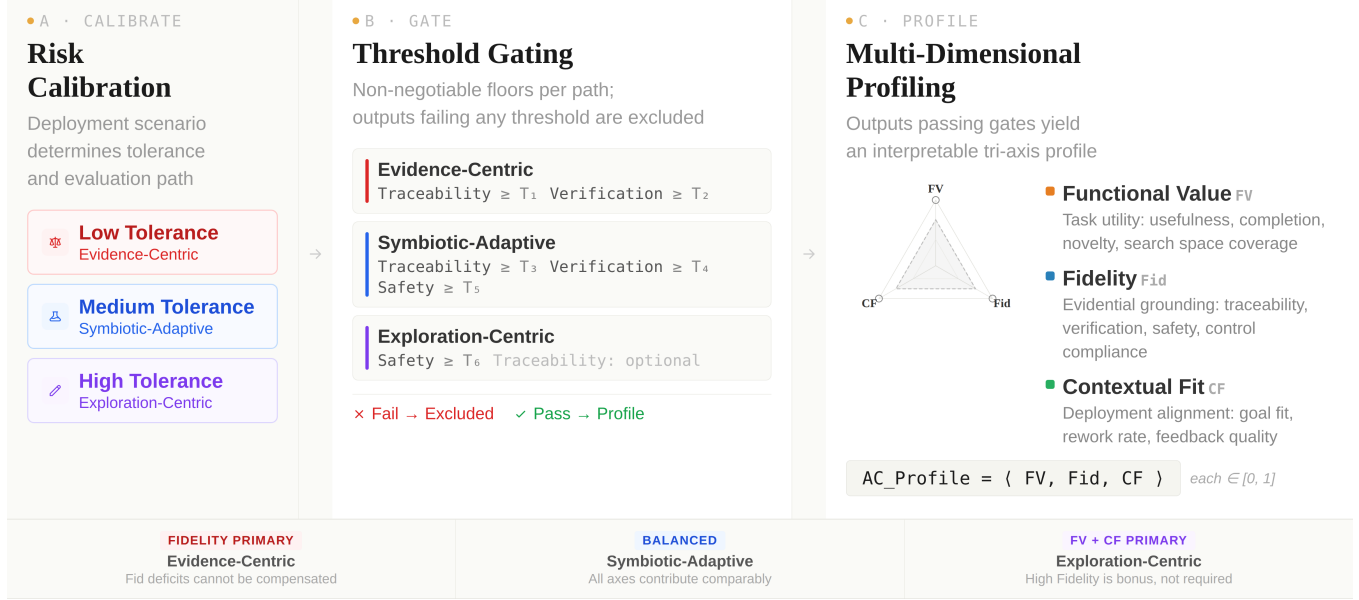


Figure 1: Overview of the Adaptive Confabulation (AC) framework. Stage A calibrates risk tolerance based on deployment context, yielding three evaluation paths. Stage B enforces non-negotiable threshold gates per path—outputs failing any gate are excluded. Stage C profiles passing outputs along three interpretable axes (Functional Value, Fidelity, Contextual Fit), with axis prioritization determined by the originating context.

Abstract

Current hallucination benchmarks impose a uniform accuracy standard that penalizes productive deviation in creative workflows while failing to rigorously gate harmful fabrication in high-stakes domains—an evaluative mismatch we term the accuracy trap. Drawing on evolutionary psychiatry’s non-pathologizing perspective, we propose the Adaptive Confabulation (AC) framework, which re-frames hallucination as a context-dependent generative cost and implements a two-stage evaluation protocol: (1) threshold gating that enforces non-negotiable floors for safety and traceability, followed by (2) multi-dimensional profiling along three axes—Functional

Value, Fidelity, and Contextual Fit—with context-appropriate prioritization. We demonstrate the framework through three documented cases spanning legal, scientific, and creative domains. Results show that AC successfully intercepts high-risk fabrication while differentiating balanced profiles suited to human-in-the-loop workflows from FV-dominant profiles appropriate for creative exploration. This work shifts hallucination evaluation from static accuracy scoring toward actionable, context-aware governance.

CCS Concepts

• **Human-centered computing** → HCI theory, concepts and models; HCI design and evaluation methods; User studies; • **Computing methodologies** → Natural language generation.

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1 Introduction

In 2023, New York attorney Steven Schwartz was sanctioned for submitting ChatGPT-generated legal filings containing fabricated case citations. Yet in the same year, researchers converted GPT-4’s speculative outputs into testable breast-cancer treatment hypotheses validated through wet-lab experiments [1]. This divergence exposes what we term the *accuracy trap*: standard benchmarks penalize productive deviation in creative workflows while failing to rigorously gate harmful fabrication in high-stakes domains. Existing taxonomies distinguish intrinsic from extrinsic hallucinations [10] or categorize knowledge conflicts [20], yet lack a procedural mechanism translating these distinctions into deployment decisions [3]. The gap is threefold: no accept/reject logic tied to risk tolerance, no operationalized metrics for functional value beyond accuracy, and no protocol preserving multidimensional information rather than collapsing into a single score.

Evolutionary psychiatry offers a reframing lens: clinical confabulation—spontaneous, non-deceptive gap-filling—represents coherence preservation under epistemic scarcity, not pathology [12, 13]. Projecting this logic onto generative models, hallucination becomes an analogous *adaptive cost*: higher sampling temperatures increase both novelty and fabrication risk [2], mirroring biological sensitivity–false-positive trade-offs. This reorientation shifts design objectives from blanket elimination to value maximization under controllable risk.

On this basis, we propose the **Adaptive Confabulation (AC)** framework, implementing threshold-gated, multi-dimensional evaluation tailored to three workflow archetypes: evidence-centric, symbiotic-adaptive, and exploration-centric. AC operates in two stages: (1) enforcing non-negotiable safety/traceability gates, and (2) generating an interpretable tri-axis profile—Functional Value, Fidelity, and Contextual Fit—with context-appropriate prioritization. We demonstrate the framework through three documented cases, validated by independent raters ($MAD = 0.034$), providing actionable governance logic for context-aware AI systems.

2 Related Work

Hallucination taxonomies and detection. Surveys of LLM hallucination have established foundational distinctions between intrinsic and extrinsic hallucination [10], between faithfulness and factuality [20], and recent benchmarks such as HalluLens [3] and HHEM [17] operationalize detection through automated pipelines. These efforts answer *whether* an output deviates from a reference, but remain silent on *what should follow*: no accept/reject logic is conditioned on deployment context, and flagged outputs receive no actionable disposition. AC inherits detection signals from such tools while adding a governance layer that translates flags into context-dependent decisions.

Multi-dimensional NLG evaluation. Moving beyond single-score accuracy, recent NLG evaluation frameworks assess outputs along multiple quality axes such as relevance, coherence, fluency, and consistency [21], while human–AI collaboration metrics extend this to task-level constructs including rework effort [5], co-creative contribution [7], and user-perceived usefulness [15]. However, these multi-axis schemes treat dimensions as uniformly weighted or leave prioritization implicit. They also lack a gating mechanism: a high

fluency score can compensate for low factuality, obscuring risk in safety-critical deployments. AC addresses both gaps through threshold gating that enforces non-compensatory floors *before* profiling, and through explicit axis prioritization calibrated to workflow type.

Context-sensitive AI governance. Within HCI, calls for context-aware evaluation have grown [6], recognizing that identical model behavior may constitute a benefit in one workflow and a hazard in another. Risk-tiered frameworks in adjacent fields, from clinical decision support to autonomous driving, demonstrate the viability of tolerance-calibrated governance, yet no analogous protocol exists for generative text outputs. AC operationalizes this principle for LLM hallucination by mapping deployment scenarios to differentiated threshold and priority configurations, providing a procedural bridge between detection-level evidence and deployment-level action.

3 Adaptive Confabulation Framework

We reconceptualize "hallucination" as *Adaptive Confabulation (AC)*: under explicit risk constraints, the model may generate content diverging from established facts yet retaining measurable task value. Evaluation shifts from "*Is it correct?*" to "*Is it useful, auditable, and appropriate?*"

3.1 Design Principles and Indicator Clusters

The framework rests on three principles. (i) **Thresholds first**: non-negotiable floors for traceability, verification, and safety must be cleared before further assessment; outputs failing any gate are excluded. (ii) **Tunable and auditable**: confabulation degree is adjusted via controllable parameters—temperature, retrieval augmentation, abstention mechanisms—while maintaining a complete audit trail. (iii) **Context-adaptive prioritization**: axis priorities are calibrated to the deployment scenario, determining which dimensions govern interpretation (Fig. ??).

Each axis is operationalized through representative metrics (Table 1). *Functional Value* captures task utility: usefulness, completion rate, novelty, search space coverage, and narrative coherence. *Fidelity* assesses evidential grounding: traceability, claim verification rate, safety, and responsiveness to control directives. *Contextual Fit* measures deployment alignment: goal fit, rework rate, and feedback interaction quality. These indicators inform the profiling stage but never override threshold gates.

3.2 Evaluation Protocol

The framework employs two sequential stages: *threshold gating* followed by *multi-dimensional profiling*.

Stage 1: Threshold Gating. Each workflow path enforces non-negotiable floors. The *evidence-centric* path requires Traceability ≥ 0.80 and Verification ≥ 0.80 . The *symbiotic-adaptive* path sets moderate floors: Traceability ≥ 0.50 , Verification ≥ 0.50 , Safety ≥ 0.10 . The *exploration-centric* path enforces only Safety ≥ 0.80 , with traceability recommended but not gating. These values are illustrative; deployment requires domain-expert calibration.

Stage 2: Multi-Dimensional Profiling. Outputs passing the gates are scored along three axes—Functional Value (FV), Fidelity (Fid),

Table 1: Tri-axis indicator clusters for AC evaluation. Axis scores are interpreted via context-appropriate prioritization rather than aggregated into a single composite.

Axis	Metric	Operationalization
Functional Value	Usefulness	Task-completion effectiveness; Likert or USE scales [11, 15].
	Task Completion Rate	Proportion of objectives successfully completed.
	Novelty Score	Inventiveness via CPSS or embedding distance [4, 18].
	Search Space Coverage	Breadth of explored solution space [8].
	Narrative Coherence	Repetition and inconsistency detection [19].
Fidelity	Traceability	Proportion of assertions with verifiable citations [19].
	Claim Verification Rate	Ratio of evidence-supported statements [16].
	Safety	Violation rate and severity of prohibited content [14].
	Responsiveness to Controls	Compliance with style, temperature, and constraint directives [6].
Contextual Fit	Contextual Goal Fit	Task-requirement alignment via satisfaction surveys [15].
	Rework Rate	Post-editing effort to refine outputs [5].
	Feedback Interaction	Efficiency of iterative improvement cycles [7].
	Quality	

and Contextual Fit (CF)—each in $[0, 1]$, producing a tri-axis profile:

$$AC_Profile = \langle FV, Fid, CF \rangle \quad (1)$$

Interpretation follows axis prioritization. In evidence-centric workflows, Fidelity is primary; deficits in FV or CF cannot compensate for inadequate grounding. In symbiotic-adaptive workflows, all axes contribute comparably; a balanced profile signals suitability for iterative human-AI collaboration. In exploration-centric workflows, Functional Value and Contextual Fit take precedence; high Fidelity is a bonus, not a requirement. This design encodes a core insight: in high-stakes contexts, dimensions should *not* compensate for one another—the tri-axis profile makes this constraint explicit. Section 4 demonstrates this protocol through preliminary expert scoring of three contrasting cases.

4 Framework Application

We apply the AC framework to three documented cases from published literature—spanning evidence-centric, symbiotic-adaptive, and exploration-centric workflows—to demonstrate how the gate-then-profile procedure differentiates contexts and reframes existing findings.

4.1 Scoring Protocol

Two independent raters scored all 12 indicators (Table 1) for each case using published documentation. Mean Absolute Difference (MAD = 0.034) and Agreement Rate (100% within ± 0.10) indicated acceptable convergence. Final axis scores (Table 2) reflect rater means; rigorous validation is outlined in Section 5.

4.2 Case Analysis

Law (Mata v. Avianca): Evidence-Centric Path. Attorney Steven Schwartz used ChatGPT to draft legal filings containing fabricated case citations, resulting in judicial sanctions. Under AC, this case fails the initial gate: Traceability scored 0.11, Claim Verification

scored 0.00—both below the required threshold (≥ 0.80). No tri-axis profile is generated; the output is mechanically excluded.

Science (Breast Cancer Hypothesis): Symbiotic-Adaptive Path. Researchers employed GPT-4 to generate breast-cancer treatment hypotheses, some validated through wet-lab experiments [1]. All thresholds were met (Traceability 0.60, Verification 0.57, Safety 0.93). The resulting profile $\langle 0.72, 0.70, 0.71 \rangle$ exhibits notable balance, indicating suitability for human-in-the-loop workflows.

Creative Writing (Wordcraft): Exploration-Centric Path. In the Wordcraft project [9], professional writers used LaMDA-based suggestions as creative stimuli. Safety (0.96) cleared the threshold (≥ 0.80). The profile $\langle 0.76, 0.86, 0.70 \rangle$ shows Fidelity (0.86) exceeding Functional Value (0.76)—reflecting high Safety and Responsiveness scores rather than factual sourcing. Under exploration-centric prioritization, Functional Value and Contextual Fit are primary; high Fidelity functions as compliance bonus.

Cross-Context Synthesis. All three cases involve questionable factual grounding, yet evaluative outcomes diverge (Table 2; Fig. 3). In law, hallucination is intercepted at the gate. In science, passage yields a balanced profile suited to iterative verification. In creative practice, an FV-dominant interpretation legitimizes deviation as feature rather than flaw. AC produces interpretable profiles whose meaning depends on deployment context.

5 Discussion and Conclusion

We position AC as a *governance protocol*—a procedural architecture that organizes existing detection signals into deployment decisions—rather than a competing detection method or a standalone theory of hallucination. This distinction addresses a gap left by automated detectors such as HalluLens [3] and HHEM [17], which flag hallucination instances but provide no guidance on whether flagged outputs should be rejected, retained, or reframed depending on context.

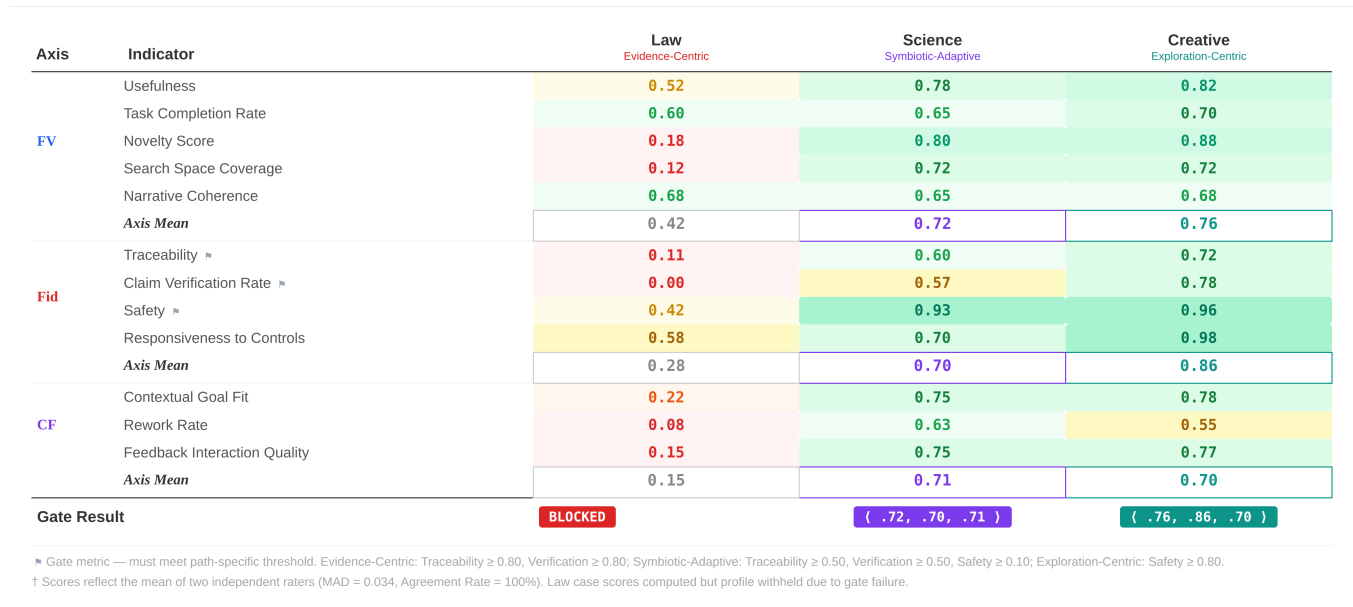


Figure 2: Full 12-indicator scoring breakdown across three cases. Color intensity reflects score magnitude. Axis means are computed via equal-weight averaging. Law case scores are computed but the profile is withheld due to gate failure. Scores: mean of two raters (MAD = 0.034, Agreement = 100%).

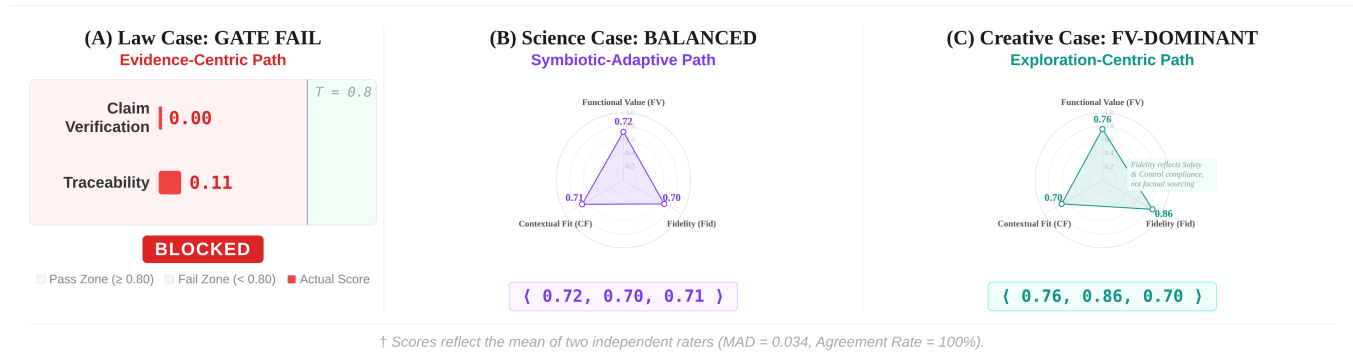


Figure 3: AC framework application. (A) Law case fails threshold gating (Traceability = 0.11, Verification = 0.00 < 0.80); no profile generated. (B) Science case yields balanced profile { .72, .70, .71 }, suited to iterative human-AI collaboration. (C) Creative case yields FV-dominant profile { .76, .86, .70 }; high Fidelity reflects safety compliance, not factual sourcing. Scores: mean of two raters (MAD = 0.034, Agreement = 100%).

Table 2: AC framework applied across three tolerance regimes. Outputs failing threshold gates are excluded from profiling. Profiles {FV, Fid, CF} preserve axis-level information; interpretation follows context-appropriate prioritization.[†]

Case	Path	Gate	Profile	Interpretation
Law (Mata v. Avianca)	Evidence-Centric	FAIL	—	Blocked: Traceability = 0.11, Verification = 0.00 < threshold.
Science (Breast Cancer)	Symbiotic-Adaptive	PASS	{ .72, .70, .71 }	Balanced; suited to human-in-the-loop iteration.
Creative (Wordcraft)	Exploration-Centric	PASS	{ .76, .86, .70 }	FV-dominant; high Fidelity is bonus, not requirement.

[†]Scores reflect the mean of two independent raters (MAD = 0.034, Agreement Rate = 100%). Full scoring breakdown available upon request.

Likewise, multi-dimensional evaluation frameworks in HCI (e.g., [7, 15]) assess human-AI collaboration quality along multiple axes yet

lack accept/reject gating tied to risk tolerance. AC bridges these layers: threshold gating operationalizes risk-calibrated accept/reject logic *upstream* of profiling, while the tri-axis scheme preserves interpretable, axis-level information *downstream* for context-appropriate decision-making.

This architecture carries direct design implications. Failed evidence centric gates can trigger submission-blocking warnings specifying deficit indicators; symbiotic-adaptive profiles can surface verification sidebars for selective human adjudication; exploration-centric workflows can foreground version control for retaining or hybridizing AI suggestions. Such profile-responsive interaction instantiates a core insight: hallucination governance requires context-sensitive calibration, not uniform suppression.

Several methodological boundaries should be made explicit. The current demonstration is *conceptual-demonstrative*: three published cases scored by two raters suffice to illustrate gate-then-profile logic and cross-context differentiation, but do not establish psychometric robustness or generalizability. Threshold values are illustrative anchors, not validated standards; operationalization in specific regulatory environments will require domain-expert calibration. The equal-weight averaging within each axis foregoes nuanced weighting that AHP or sensitivity analysis might provide.

These boundaries define a concrete validation agenda: (1) domain-specific expert panels ($n \geq 6$) enabling formal inter-rater reliability via ICC or Krippendorff's α ; (2) threshold sensitivity analysis establishing empirically grounded parameter ranges across regulatory regimes; and (3) longitudinal field deployments tracing effects on trust calibration, rework costs, and risk spillover.

In summary, the Adaptive Confabulation framework reconceptualizes hallucination evaluation from accuracy-centric scoring to context-aware governance. By implementing threshold gating followed by tri-axis profiling, AC produces interpretable outputs supporting nuanced decision-making across divergent workflows. Progress lies not in eliminating hallucination but in containing it—channeling generative variability into controlled zones through thresholds, traceability, and context-appropriate interpretation.

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